INPUT cross-section · lay-ups · length L · end conditions · load n_0

A CROSS-SECTION ANALYSIS

once per cross-section, reused for every length (Sections 2.1, 2.2)

1 Discretize the cell

walls → 93 segments, 92 nodes

nodal $\boldsymbol{U}$, $\boldsymbol{\Theta}$, warping (Eq. (8))



2 Mode families, side edges free

differs from classical GBT

G D L T conventional

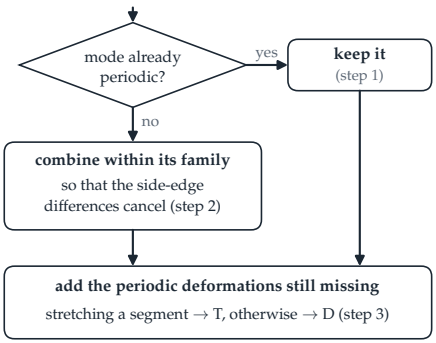
S M global, distortional, local, transv. extension · shear, Mindlin (Section 2.1.3)

3 Make every single mode periodic

NEW equal values at s_L and s_R, mode by mode (Proposition 1)

G D L T in-plane modes, family by family

periodic condition: $\boldsymbol{R}\,[\boldsymbol{U};\boldsymbol{\Theta}] = \boldsymbol{0}$ (3 equations, Eq. (12))



⇒ 273 modes = $3N - 3$: nothing lost, nothing spurious

G D warping u_k

closed contour

$u_k(s_L) = u_k(s_R)$:
the cell is a contour closed through its side edges

least shear strain energy
Eq. (9) → linear system
Eq. (13)

open part: $\gamma_k = 0$ (Vlasov)
closed part: shear flow kept

⇒ unique periodic warping

S M nodal functions

one function per node



merge the two side-node functions
into one, Eq. (14)

⇒ 91 + 205 modes

(axial mode in S)

Periodic basis: 569 modes

every mode, every truncated set and every Ritz trial function is periodic

4 Stiffness tensors

cross-section integrals of the modes, Eqs. (17)–(26)

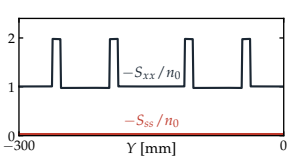


B MEMBER ANALYSIS

repeated for every length L and end condition (Section 2.3)

5 Pre-buckling state

computed, not assumed



linear solution under n_0, Eqs. (30), (31)

6 Geometric tensors

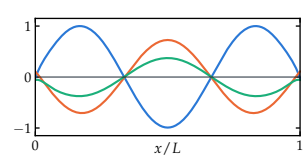
second-order work of S_{xx}, S_{ss}, S_{xs} on the buckling mode

S_{ss} comes from the constant cell width (Section 3.2)

Eqs. (27)–(29)

7 Ritz solution

amplitudes $\phi_k(x)$ of modes 12, 18, 8



B-splines · $(\boldsymbol{K} + \lambda\,\boldsymbol{G})\,\boldsymbol{a} = \boldsymbol{0}$, Eqs. (37), (38)

pictures: computed, Unsym cell, $b_s = 150$ mm, $\beta = 1.5$, SS

new in this study

tag differs from classical GBT

OUTPUT N_{cr} (32.26 N/mm here) · mode · family participations (Eq. (41))